

Research Paper

Emerging energy economics and policy research priorities for enabling the electric vehicle sector

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ABSTRACT

We highlight key emerging research priorities on the climate- and industrial-policy driven pathway to passenger car electrification, including both pure battery electric and plug-in hybrid electric vehicles, using a three-pronged approach. It includes examining journalistic reports from a research perspective, corroborating the determined research priorities with recent literature, and consulting a diverse group of experts in policy, technology, fuel and infrastructure fields. We delineate eight overarching themes that correspond to distinct research challenges and prospects and rank them by integrating expert ratings on importance in enabling the EV sector, novelty and feasibility using a multi-criteria decision-making (MCDM) framework. Specific emerging research issues related to – (i) assessing the impact of U.S. and European policies on the competitiveness of their domestic electric vehicle (EV) battery industry in relation to China and global trade in batteries; (ii) exploring the prospect and effect of rising metal prices on the battery and EV markets; (iii) assessing the EV price trends and the continued necessity of government subsidies in light of declining battery costs and EV price wars among automakers – ranked among the top three. The sensitivity analysis conducted within the MCDM framework, utilizing three distinct response weighting schemes, demonstrated greater variability in expert opinions-based ranking for two issues compared to others: (i) the geopolitical battle for control of critical metals for EVs, and (ii) price war and competition in the EV market. Pursuing this new research agenda should help inform policymakers, and industry decision makers to ensure a smooth ride towards passenger car electrification.

1. Introduction

A number of nations are pushing ahead on a path toward passenger car electrification as part of their climate, environmental and industrial policies, including the aspiration of dominating the global automotive market (Meckling and Nahm, 2019; Dua, 2024). The European Union, for instance, has enacted legislation aiming for a 100 % share of new vehicle sales with zero tailpipe emissions, particularly electric vehicles, barring an exemption permitting internal combustion engine vehicles (ICEVs) running on carbon-neutral synthetic fuels (Hancock et al., 2023; Council of the European Union, 2023). California mirrors this approach, mandating the exclusive sale of zero tailpipe emission vehicles by 2035 (Bushey and Grimes, 2022; California Air Resources Board, 2022).

China's New Energy Vehicle policy sets escalating targets for EVs and other new energy vehicles (International Council on Clean Transportation, 2021). In the United States, increasingly stringent fuel economy and CO₂ standards favor EV adoption (Bushey and Williams, 2023), complemented by demand-side incentives such as federal tax credits under the 2022 enacted Inflation Reduction Act (IRA) (Laing, 2023). These measures reflect not only climate concerns but also industrial ambitions (Dua, 2024). China, for example, views EVs as an avenue to establish a strong foothold in the global automotive market, traditionally dominated by automakers from Japan, Germany, and the United States (U.S.), which specialize in ICEV manufacturing (White and Campbell, 2023; Liu et al., 2020; Campbell et al., 2023). In addition to being driven by climate and industrial policies, the shift towards

Abbreviations: EV, Electric Vehicle; ICEV, Internal combustion engine vehicle; CARB, California Air Resources Board; ICCT, International Council on Clean Transportation; IRA, Inflation Reduction Act; US, United States; EU, European Union; ZEV, Zero Emission Vehicle; OPEC, Organization of the Petroleum Exporting Countries; MCDM, Multi-Criteria Decision Making; AHP, Analytic Hierarchy Process; TOPSIS, Technique for Order Preference by Similarity to the Ideal Solution; CR, Consistency ratio; CI, Consistency index; FT, Financial Times.

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passenger car electrification forms a critical component of several nations' environmental strategies. This transition recognizes the imperative to mitigate environmental pollution by reducing not only direct tailpipe emissions but also by addressing the broader lifecycle emissions associated with vehicle use. Moreover, the development of effective EV policies necessitates a holistic environmental evaluation that extends beyond immediate operational emissions to consider the full environmental footprint of vehicles, from manufacture through disposal, thereby ensuring a truly sustainable transition towards electrification. The culmination of these climate, environmental and industrial policies is anticipated to significantly boost demand for EVs. In the pursuit of achieving their EV aspirations, particularly amidst rising geopolitical tensions, it is imperative for nations to recognize emerging research challenges so that they can strategically allocate resources.

We identify such emerging research challenges along the passenger car electrification pathway in this work. The core research questions we seek to address are: 1) What are some of the contemporary thematic areas related to passenger car electrification being reported in the media? and 2) What are some of the policy and economic research priorities that emerge from these contemporary issues? These questions guide our exploration of various aspects of the EV sector, aiming to delineate an emerging policy and economic research agenda addressing which can facilitate further development of the EV sector.

This is, as far as we are aware, the first attempt at prioritizing emerging research challenges in EV sector by analyzing recent journalistic investigations through an economics and policy research lens, supported by academic literature, and expert consultations. The identified research priorities could provide valuable guidance for future academic and industrial research endeavors. These efforts have the potential to yield actionable recommendations for policymakers and industry stakeholders. Decision-makers at research funding agencies would also potentially derive value from these findings when it comes to the prioritization of future research grants.

In terms of research gaps that this work addresses, it is worth acknowledging that the formulation of future research agendas is typically confined to either brief paragraphs in the concluding sections of academic papers or is often narrow enough to be concentrated on the specific sub-theme explored within that paper. In many cases, future research directions are primarily derived from the authors' expert judgment and frequently lack validation from outside experts. Furthermore, there is a lack of emphasis on the alignment between the outlined future academic research priorities and the perspectives of non-academic stakeholders regarding the research that they believe is necessary to tackle the contemporary challenges encountered by the sector. This work addresses these gaps by delineating a wide range of emerging issues that warrant deeper investigation identified through journalistic reports that include views of non-academic stakeholders in corroboration with academic literature and prioritized through consultation with a diverse group of energy economics and policy experts.

In recent years, there has been a notable surge in scholarly publications dedicated to formulating research priorities (Luè et al. 2016; Azad and Parvin 2022; Serdeczny and Lissner, 2023; Basak et al., 2019; Parag et al., 2023; Chen et al., 2022), reflecting a growing recognition of the importance of strategic and forward-looking scholarly inquiries, including some in the passenger car electrification domain. Some studies delve into the factors that significantly influence consumer purchasing choices for EVs, comparing these influences across academic and practitioner publications to forge a consolidated research agenda on sustainable mobility (Brescia et al., 2023). Singh et al. (2023) examine the prevailing research trends and future agendas concerning consumer EV adoption within clean energy initiatives. Similarly, the work of Rezvani et al. (2015) addresses the drivers and barriers to consumer adoption of EVs, seeking to enhance our theoretical understanding of consumer intentions and behaviors, while identifying critical research gaps and future directions. Liao et al. (2017) focus on how consumer preferences for EVs can inform policymaking and guide subsequent research efforts.

Secinaro et al. (2022) highlight the behavioral factors that influence consumer decisions to purchase EVs and call for further research to understand and enhance these behaviors. Bose Styczynski and Hughes (2019) explore how governments in leading markets utilize policy instruments to promote personal transport electrification, questioning whether these policies are technology-neutral across different types of EVs and suggesting a research agenda to understand policy uniformity. Aksen et al. (2022) investigate the effectiveness of zero-emission vehicle (ZEV) sales mandates in driving long-term sales and their contribution to the deep decarbonization of the transport sector, while identifying the need for further research to understand their role in an effective policy mix. Amedokpo and Boutueil (2023) propose a research agenda for EVs in Africa, considering their role both as a subject of academic research and as a practical alternative to internal combustion engine vehicles. Lastly, Sovacool et al. (2022) scrutinize the extent to which low-carbon technologies, including EVs, are equitable and the potential policy and research implications of their associated behaviors moving forward.

The present study distinguishes itself by proposing a comprehensive agenda for policy and economic research aimed at enabling the EV sector. Unlike many studies that primarily rely on literature reviews, this research employs a three-pronged approach, which includes examining journalistic reports using a research lens, comparing the determined research priorities with recent literature, and consulting a diverse group of experts. This methodology not only differentiates it from previous work but also enriches the research agenda by integrating diverse perspectives from various stakeholders involved in the EV sector, including those from both inside and outside the academic community.

2. Background

We provide a summary of the relevant grey literature and peer-reviewed journal articles, clustered by themes.¹ In particular, each theme includes a concise analysis of the relevant media (news) articles. In addition, we provide specific examples of relevant research questions that have been investigated so far in academic literature in relation to the emerging issues. The derived condensed and organized themes could serve as a valuable resource for individuals seeking a concise and broad overview of the emerging EV sector discourse.

Theme 1: Governments' Strategies to Localize the EV Battery Supply Chain

In such media (news) inquiries, the Western nations' strategies and challenges related to diversifying EV battery supply chains away from China, particularly middle and downstream segments, are among the frequently discussed topics (Bounds, 2022; White and Lewis, 2022; Beattie, 2022). This issue encompasses the high global market share held by Chinese groups in the production capacity for battery electrodes (Bounds, 2022). There is a focus on the efforts of the U.S. and Europe to decrease their reliance on China for EV batteries through significant capital expenditure and strategic initiatives (White and Lewis, 2022). The theme also involves understanding the geopolitical and economic implications of China's dominant position in the global battery supply chain, as well as the competition between China and Western nations (Beattie, 2022).

In the realm of academic research, a significant focus has been placed on understanding the feasibility of the critical mineral targets set by the U.S. Inflation Reduction Act (IRA) for EV batteries. This exploration, as

¹ The themes were identified through a systematic grouping of media (news) articles based on thematic content. The methods section provides a detailed explanation of how the themes are derived. However, it is important to acknowledge that the themes identified in this analysis do not encompass all issues related to passenger electrification. There may be blind spots or specific issues that have not yet been reported in the media, and thus remain unaddressed in this work.

highlighted by [Trost and Dunn \(2023\)](#), not only considers different battery types but also delves into the environmental and supply chain implications associated with these targets. Another critical area of investigation, as noted by [Vivoda \(2023\)](#), involves examining the concept of "friend-shoring."² This strategy is particularly relevant for Western nations seeking to mitigate the vulnerabilities in their supply chains, especially concerning the critical minerals essential for clean energy technologies. Furthermore, scholars like [Kleimann et al. \(2023\)](#) have been exploring the main differences between the U.S. IRA and its European counterparts. This comparison primarily revolves around the aspects of green subsidies and their broader implications on the international trading system and future green industrial policies. Lastly, the research by [Bown \(2023\)](#) sheds light on the implications of the U.S. IRA on transatlantic cooperation, particularly between the U.S. and the EU, considering the shared concerns about various economic, social, and geopolitical factors.

Theme 2: EV Rollout Challenges: Rising Raw Material Prices and Automakers' Responses

The major common issue being explored in this theme is the rising cost of lithium-ion batteries in 2022 and its impact on the EV industry. With lithium-ion battery prices increasing for the first time in 2022 over a decade due to rising raw material costs ([Dempsey, 2022](#)), car manufacturers are seeking ways to secure a stable and affordable supply of batteries. This includes carmakers investing in early-stage mining companies as well as forging direct deals with the mining sector ([Bushey and Dempsey, 2022](#)).

Pertinent questions explored in recent academic literature include [Wang et al.'s \(2023\)](#) investigation into the complexities faced by China in balancing its ambitious EV and climate objectives with the rising prices of critical battery metals, scrutinizing the influence of price volatility on the competitiveness and expansion of the EV market. Additionally, European automakers' innovative strategies in adopting circular business models to navigate supply constraints related to lithium-ion batteries have been examined by [Albertsen et al. \(2021\)](#). Concurrently, the research by [Yang and Fulton \(2023\)](#) provides insights into the evolving landscape of investments and production capacities in the battery and EV sectors. Their work critically evaluates the existing and potential gaps between the burgeoning production capabilities and the aggressive climate-driven targets, underscoring the need for strategic planning and investment in this rapidly evolving industry.

Theme 3: The Geopolitical Battle for Control of Critical Metals for EVs

Predominantly, media (news) reports on this topic concentrate on the efforts of resource-rich nations to collaborate in order to exert enhanced unified control over the supply of EV battery metals. They also discuss the economic and strategic implications of such cooperative measures. This includes exploring the geopolitical dynamics surrounding nations rich in these resources and their efforts to control global supply and pricing, as well as the potential formation of Organization of the Petroleum Exporting Countries (OPEC)-like groups for critical battery metals ([Dempsey and Ruehl, 2022](#)). The reports highlight Indonesia's consideration of forming a nickel group, as well as previous discussions between Chile, Argentina, and Bolivia about forming a similar lithium group ([Dempsey and Ruehl, 2022](#)). The inquiries also address such countries' approaches to potential foreign partnerships,

such as Bolivia's approach to potential foreign partnerships for its state lithium company, as well as the principles that guide such agreements to ensure benefits for their people ([Graham, 2022](#)).

Key academic inquiries in this realm include an examination by [Mancheri \(2015\)](#) into the implications of Chinese export restrictions on the global trade of "critical" materials, with a specific focus on rare earths. This research delves into the consequential impacts of such restrictions on the accessibility of these vital materials for industrialized nations, highlighting the intricate interplay between national policies and global supply chains. Complementing this perspective, [Månberger and Johansson \(2019\)](#) explore the geopolitical significance of metals that are essential for the green transition, particularly in the context of renewable energy. Their investigation encompasses various dimensions such as the concentration of resources, potential revenues for resource-rich nations, and the overarching global market dynamics.

Theme 4: Battery Start-ups and Government Support

In this set of media enquiries, the role of government support in developing domestic battery gigafactories is the central theme. This includes focus on how government backing can lead to the establishment of such gigafactories, which are expected to create jobs and contribute to the growth of the battery industry ([Dempsey and Williams, 2022](#)). Additionally, there is an emphasis on the importance of investor confidence and how it can impact the cost-effectiveness of building these gigafactories with potential implications for public spending ([Rutter Pooley, 2022](#)).

Research by [Ram et al. \(2022\)](#) sheds light on the employment and economic ramifications of a global energy shift. Their work critically evaluates the roles of various sectors and value chains, with a special emphasis on electric vehicle batteries, and contrasts these with the traditional fossil fuel industry. Another significant strand of research, led by [Surana et al. \(2023\)](#), focuses on the role of public policy in fostering climate-tech start-ups. Their investigation centers on how government policies can not only encourage but also ensure responsible private investments for the advancement of sustainable technology and long-term decarbonization. Furthermore, the work of [Doblinger, Surana, and Anadon \(2019\)](#) delves into the direct interactions between governments and start-ups, particularly in the cleantech sector. Their study explores the influence of these interactions on various facets such as innovation, technological development, and the mobilization of private sector investments.

Theme 5: EV Pricing and Government Incentives: New and Used Vehicle Markets

This theme focuses on the affordability of electric cars. It involves the concern that while electric car prices are expected to decrease with declining battery costs, the current list prices of most electric cars are still relatively high ([Campbell 2022d](#)). Additionally, there is a discussion about the impact of removing exemptions from annual car tax on the affordability of electric vehicles ([Campbell 2022d](#)). Some argue that this change could make electric cars less affordable, especially in the second-hand market, while others see it as a measure to make the motoring tax system fairer ([Campbell 2022d](#)).

In terms of academic inquiry, [Goetzel and Hasanuzzaman \(2022\)](#) delve into the trends in EV retail prices, probing into the critical juncture at which EVs may achieve purchase price parity with conventional vehicles across diverse market segments. Complementing this, [Tankou, Lutsey, and Hall \(2021\)](#) have extended the scope of research to encompass the used ZEV market. Their work sheds light on the current trends, technological advancements, and policy implications that are vital for comprehending and facilitating the growth of the used ZEV sector. This includes an in-depth analysis of the dynamics of pricing, durability of these vehicles, the role of incentives, and the requisite charging infrastructure.

² "Friend-shoring" refers to the practice of strategically diversifying supply chains by relying more on allies and friendly nations for critical resources, aiming to reduce dependence on potentially adversarial or unreliable sources. It involves fostering closer economic relationships with trusted partners to enhance supply chain resilience. This term is used in the context of supply chain security and geopolitical considerations.

Theme 6: Challenges and Setbacks for EV Start-ups

Media investigations in this theme center on the challenges and setbacks faced by EV start-up companies. These challenges include issues related to capital investments, supply chain disruptions, and financial difficulties, which can impact their ability to meet production targets, form partnerships, and access expected financial reserves (Campbell and Nilsson, 2022; Campbell 2022b; Bushey, 2022).

A key domain of scholarly inquiry pertains to the strategies that can enable these start-ups to penetrate the automotive market effectively. Thomas and Maine's (2019) study on Tesla Motors offers crucial insights into such market entry strategies, underscoring Tesla's journey as a case study for emerging enterprises in this domain. Additionally, there is a focus on how start-ups in the electric vehicle industry are crafting innovative business models to establish a dominant market position, as explored by Donada and Lepoutre (2016).

Theme 7: Price War and Competition in the EV Market

The central theme here revolves around the competitive landscape, market entry strategies, and pricing dynamics in the global EV market, with a particular focus on the roles of Chinese and European automakers. This competition includes "price war" in China in response to its softening EV demand within an increasingly crowded EV sector (Li and Nilsson, 2022), as well as the entry of Chinese car manufacturers like BYD, Great Wall, Nio, Geely, and SAIC Motor into the European market with electric and hybrid vehicle offerings (Campbell 2022a; Kastner, 2022). Executives of European automakers are additionally concerned regarding the pricing tactics employed by Chinese automakers, who may be inclined to offer vehicles at a loss in order to secure a portion of the European market (Campbell 2022a). In an effort to prevent unfair competition, these executives urge the European Union to contemplate protectionist measures to safeguard local manufacturing by considering the implementation of tariffs on automobiles imported from China (Campbell 2022a).

A pivotal area of academic investigation, as noted by Bown (2023), revolves around the impact of protectionist measures, exemplified by the U.S. Inflation Reduction Act, on the EV supply chain. This inquiry not only unravels the intricate political-economic challenges that emerge in the context of US-EU cooperation on clean energy transition policies but also brings to the fore the critical issue of balancing these regional strategies against the backdrop of China's burgeoning dominance in the sector Bown (2023).

Theme 8: Emerging Technologies in EV Battery and Charging Solutions

Such reporting encompasses concerns related to the financial viability and progress of companies developing new EV technologies, including solid-state batteries and software for electric car chargers. It highlights the challenges faced by investors when companies, initially hyped and well-funded, fail to deliver on their promises, leading to significant declines in market value (Financial Times, 2022). Additionally, it touches upon the importance of technological advancements in the EV sector, such as software that enhances the efficiency and interaction of electric car chargers with both vehicles and the electrical grid (Campbell 2022c).

Recent academic inquiries, including the work of Chen et al. (2020), are centered on elucidating the challenges, mechanisms, and design strategies essential for the development of solid-state batteries that are both practically accessible and stable. Studies, including those by Sadeghian et al. (2022), provide a comprehensive exploration of the multifaceted aspects of smart charging for EVs. Their investigation focuses on identifying effective solutions, strategies, and technologies that not only meet the charging demands of EVs but also mitigate their impact on the power grid.

3. Methods

In this work, we identify key research priorities for enabling the EV sector. Specifically, we first derive overarching themes by summarizing and categorizing the wide-ranging scope of recent media investigations pertaining to the EV sector. The emerging themes were formulated by performing a theme-based grouping of recent EV reporting from October 2022 to December 2022 in the Financial Times (FT). We analyzed all 77 FT articles published under their "Electric vehicles" topic over a three-month period. The theme-based grouping continued until an examination of the latest 10 articles revealed no new distinct themes. Specifically, we converted each article into a question highlighting the gist of topic explored and grouped the questions into categories based on their common themes. We began the grouping exercise with two completely different questions as seeds for two unique themes. The remaining questions were then assigned to one of the existing themes or a new theme based on whether they explored similar topics. Priority was given to incorporating the question into an existing theme. The scope of the existing themes was expanded if necessary to accommodate a new question. If a question could not be assigned to an existing theme, even after scope extension was considered, the question was assigned to a new theme. The procedure was repeated until all questions were assigned to different themes.³ Given the global nature of the EV sector, FT's global coverage, and our focus on questions rather than findings, we use FT to sample recent media investigation themes. Since our goal for this grouping exercise is to identify key themes by extracting a wide range of questions, the FT articles are considered to provide a reasonable overview.

We start off with media reports as they frequently shed light on nascent energy economics, policy, and sustainability issues before they are thoroughly examined in scholarly literature. This can be attributed to the delays inherent in securing research funding and the time-consuming peer-review process that characterizes academic publishing. Furthermore, media reports frequently incorporate perspectives from non-academic stakeholders, often underrepresented in scholarly articles. Finally, we delineate specific research challenges and priorities that require more thorough consideration by applying an energy economics and policy research lens (including academic and industrial) on the formulated emerging themes.

We solicit expert opinions to prioritize the identified research challenges. In particular, an electronic survey was conducted by sending invitations to experts between June and July 2023.⁴ Participants in the survey came from think tanks and academia, including University of California, Davis, USA; Chalmers University, Sweden; Indian Institute of Management (IIM), India; Rocky Mountain Institute (RMI), USA; Gulf Research Center, KSA; Ajou University, Korea; Birla Institute of Technology & Science (BITS) Pilani, India; University of Maryland, USA, among others. The participants were asked to evaluate the different research challenges and opportunities outlined in the preceding sections using a Likert-type (one to five) scale, assessing their importance, novelty, and feasibility. The respondents were also asked to rate the

³ It is worth noting that some questions appeared to fit into multiple categories. Given the qualitative nature of this particular grouping exercise and the purpose of this study, the authors' expert judgment was used to group such questions under a particular theme.

⁴ The experts were selected based on their extensive experience in the policy and economic dimensions of passenger car electrification research. Moreover, the views of non-academic stakeholders were already substantially represented, as the research questions were based on contemporary issues frequently reported in the media, where non-academic perspectives are often highlighted. Consequently, there was a strategic focus on engaging academic stakeholders more intensively. Invitations were also extended to experts from policymaking institutes. Some of these experts did respond but requested that their identities and the names of their institutions remain anonymous; therefore, these details have not been disclosed.

significance of each criterion, i.e., importance, novelty, and feasibility, relative to one another. Furthermore, data was gathered pertaining to the professionals' affiliations with either non-profit or for-profit entities, their educational credentials, years of experience in the EV sector, and their level of familiarity with the research subjects under examination in this survey.

The expert survey results were analyzed using a multi-criteria decision making (MCDM) framework. AHP (Analytic Hierarchy Process) (Saaty, 1988) was used to calculate the weight scores for the three criteria – importance, novelty and feasibility – by having respondents make a series of pairwise comparisons. TOPSIS (Technique for Order Preference by Similarity to the Ideal Solution) (Hwang et al., 1981) was used to rank the different alternatives based on their normalized scores – a measure of the extent of similarity between alternatives and the hypothetical ideal solution (with importance, novelty and feasibility ratings of five each) - considering AHP-based criteria weights. While other equally or even more capable tools are available, we opted for the AHP and TOPSIS methods due to their simplicity, effectiveness, and wider acceptability. AHP facilitated straightforward pairwise comparisons between the three criteria, significantly reducing the cognitive load on respondents and streamlining the data collection process. Similarly, TOPSIS focused on identifying solutions that maximize benefits while minimizing drawbacks, thereby facilitating effective decision-making. These methods were particularly suitable given the constraints on time and the need for straightforward data collection from respondents. Given the subjective nature of the expert survey, it would have been better to use AHP for both deriving weights for criteria and ranking alternatives. From a time perspective, however, requiring respondents to make pairwise comparisons between twelve alternatives appeared impractical. Finally, a sensitivity analysis was performed using three different schemes to calculate the weighted average of the ratings given by the experts for use in the TOPSIS framework, including: (i) equal weighting, (ii) experience-based weighting, and (iii) familiarity-based weighting.

The TOPSIS process includes (Fig. 1) (Ramya and Devadas, 2019; Fancello et al., 2019; Rane et al., 2023; Heidari et al., 2022):

1. Constructing decision matrix $(X_{mn})_{ij}$ containing i alternatives and j criteria, with the integration of each criterion and alternative given as X_{mn}
2. Using normalization method to normalize the decision matrix

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}}, \quad i = 1, 2, 3, \dots, m; j = 1, 2, 3, \dots, n \quad (1)$$

3. Constructing the weighted normalized matrix by assigning a weight vector to the criteria group, where the criteria weights are determined using the AHP approach as outlined below.

$$v_{ij} = w_j r_{ij}, \quad i = 1, 2, 3, \dots, m; j = 1, 2, 3, \dots, n \quad (2)$$

4. Determine positive (A^+) and negative ideal solution (A^-)

$$A^+ = \{v_1^+, \dots, v_n^+\}, \quad v^+ = \{\max(v_{ij}), j \in J; \min(v_{ij}), j \in J'\} \quad (3)$$

$$A^- = \{v_1^-, \dots, v_n^-\}, \quad v^- = \{\min(v_{ij}), j \in J; \max(v_{ij}), j \in J'\} \quad (4)$$

5. Calculate the separation measure from both positive (S_i^+) and negative (S_i^-) ideal alternative.

$$S_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^+)^2}, \quad i = 1, 2, \dots, m; \quad (5)$$

$$S_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^-)^2}, \quad i = 1, 2, \dots, m; \quad (6)$$

6. Relative closeness to the ideal solution is calculated.

$$C_i^* = \frac{S_i^-}{S_i^+ + S_i^-}, \quad 0 < C_i^* < 1 \quad (7)$$

7. Using normalization method to normalize the closeness to the ideal solution score

$$norm_C_i^* = \frac{C_i^*}{\sum_{i=1}^m C_i^*} \quad (8)$$

8. Preference order is ranked, the larger the value, the better would be the performance of the alternative.

As described previously, the relative weights of criteria are determined using AHP, which employs ratio scale factors for pairwise comparison (Ruiz et al., 2020; Saaty, 1996, 2001; Marzouk et al., 2023; Khojastehmehr et al., 2019; Manirambona et al., 2022; Albert et al., 2016; Carrese et al., 2022; Jbahi et al., 2022). In deploying the AHP methodology, we selected three criteria central to our analysis: importance, novelty, and feasibility. As highlighted previously, 'importance' refers to the degree to which the research challenge is pivotal for

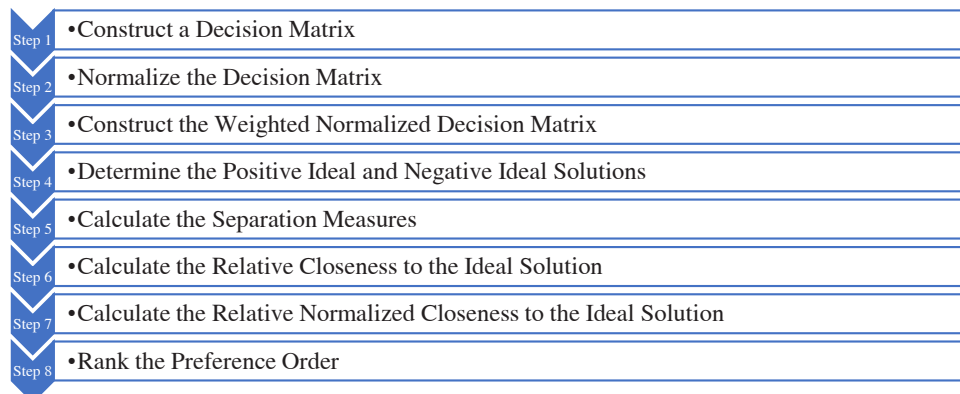


Fig. 1. Steps in the TOPSIS process.

advancing the EV sector, representing the potential impact or utility of addressing the research question. ‘Novelty,’ on the other hand, gauges the originality of the research challenge and its capacity to contribute new knowledge to the field. Finally, ‘feasibility’ assesses the ability to conduct research on a topic considering constraints in terms of time, cost, logistical challenges, data, resources, and so on. The selection of these criteria was motivated by the need to balance ambition with practicality and to ensure that the research priorities identified would not only drive forward academic inquiry but would also be actionable within the current state of the sector. This balanced emphasis on importance, novelty, and feasibility ensures a comprehensive, yet pragmatic, prioritization of emerging research challenges in the EV sector. The AHP process facilitated a structured approach to quantifying expert opinions on these dimensions, enabling a systematic derivation of weighted criteria that reflect the collective judgment of the expert panel. Specifically, we employ a pairwise comparison matrix for three criteria with pair relative significance ranks determined by a 9-point Likert scale, from which the relative criteria weights were derived using the standard AHP priority-matrix normalization method (Saaty, 1996) and the AHP priority calculator introduced by Goepel and Klaus, 2018. To ensure the integrity and robustness of our AHP analysis, a thorough examination was conducted on the consistency of respondents’ pairwise comparisons. The consistency ratio (CR) was calculated following Saaty’s (1988) method, providing a measure for evaluating the reliability of the pairwise judgments. A CR below 0.1 is deemed acceptable, suggesting that the judgments are sufficiently consistent. In order to check the consistency of the pairwise scores, we calculated the consistency ratio, $CR = CI/RI$, where the consistency index (CI) is defined by:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \tag{9}$$

with $\lambda_{\max}$ the eigenvalue of the pairwise comparison matrix and n the criteria number, where the random consistency index values, RI , for the n values have been considered from (Saaty, 2001).

4. Results & discussion

In this section, we explicitly outline the specific research challenges that demand further in-depth examination achieved by applying a comprehensive energy economics and policy research perspective, encompassing both academic and industrial viewpoints, to the identified emerging themes. Following that, we present the results of the expert survey assessment that prioritize the identified research challenges.

4.1. Emerging research challenges

In this section, we identify and discuss the emerging research challenges that stem from the contemporary issues highlighted previously. Additionally, we explore why these issues constitute ‘research challenges,’ aiming to tease out the underlying causes of these challenges.

What are they?	Why Do They Pose a Challenge?
Theme 1: Governments’ Strategies to Localize the EV Battery Supply Chain A pivotal area of future research is the assessment of the effectiveness of U.S. and European policies. This investigation is critical in understanding their role in diminishing dependence on external sources, particularly China, for EV batteries. Such an analysis not only provides insights into the success of these policies but also illuminates their broader implications. This includes the impact on domestic industry growth, where the focus	As these strategies have only recently been introduced, their real-world impact is yet to be fully assessed. However, it is feasible to explore their potential effects within a simulated environment, which can provide preliminary insights into their effectiveness and potential repercussions. For instance, gathering and analyzing expert opinions can assist in calibrating parameters of simulation studies focused on predicting the impacts of these policies. These approaches can help overcome the challenges posed by the lack of long-term empirical data and the evolving nature of policy frameworks.

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What are they?	Why Do They Pose a Challenge?
would be on evaluating how these strategies foster local industry expansion and enhance competitiveness in a global market. Additionally, it is imperative to examine the repercussions these policies may have on international trade relations.	
Theme 2: EV Rollout Challenges: Rising Raw Material Prices and Automakers’ Responses A paramount emerging research challenge lies in understanding how fluctuations in the prices of raw materials could significantly influence the production costs of batteries and EVs, thereby impacting EV sales dynamics. Another area ripe for investigation is the effect of car and battery manufacturers’ investment strategies and partnerships throughout the supply chain. By examining the intricate web of supply chain dependencies and partnerships, researchers can shed light on how these collaborative efforts might influence the economics of raw material procurement	Few companies systematically track the prices of raw materials, production costs of batteries, or the prices of EVs and that too at an aggregate level. This scarcity of data stems primarily from the confidentiality maintained by raw material producers, battery manufacturers, and EV producers regarding their cost structures. Additionally, even when datasets are available, they are often prohibitively expensive and compiled by different data providers, making integration and analysis challenging. Consequently, research in this area has been limited. However, conducting expert elicitation studies and integrating resulting insights into scenario-based simulation studies could represent one approach to examine the impact of changing EV prices on sales dynamics.
Theme 3: The Geopolitical Battle for Control of Critical Metals for EVs An emerging research challenge is the need for a comprehensive cost-benefit analysis of the formation of an OPEC-like group for battery metals. In addition to examining the consequences for supply chain stability and market prices, such an analysis could also contemplate the possible alteration in market power dynamics. Concurrently, there is a pressing need to understand the impact of the surging demand for critical metals on the economic growth and social welfare of supplier countries.	Lack of comprehensive data on the supply chain dynamics and market behaviors specific to critical metals used in EVs complicates the task of predicting market responses to changes, such as the formation of an OPEC-like group for battery metals. Additionally, many regions projected to supply critical metals in the future have historically experienced significant geopolitical instability, leading to insufficient data collection and research focus in these areas. As a result, the countries concerned may currently lack detailed and reliable economic models capable of adequately answering such questions.
Theme 4: Battery Start-ups and Government Support A pivotal emerging research challenge lies in understanding the impact of government support in attracting investments for supporting the development and competitiveness of the domestic battery industry. This issue intertwines with the broader context of economic strategy, where a cost-benefit analysis of government support for domestic battery industry becomes imperative, including implications for the long-term economic growth potential.	Given the recent emergence of government support for battery startups, studies in this field should be capable of utilizing panel data over shorter timeframes. Additionally, it is important to assess the extent of local and federal government support received by different battery startups at various times, alongside the investment these companies have secured from various sources including the private sector. Datasets covering these aspects are currently being compiled by only a few private firms, which typically charge significantly for this type of information. Alternatively, significant research effort must be expended to independently collate such data. This scarcity of accessible and comprehensive data presents a substantial challenge, complicating efforts to analyze the impact and efficiency of government initiatives on the burgeoning battery industry.
Theme 5: EV Pricing and Government Incentives: New and Used Vehicle Markets An emerging research challenge lies in assessing whether the prices of electric cars have decreased over time, particularly in light of the	To answer such questions, comprehensive datasets are required concerning the costs of EV models, including specific battery types and battery costs. However, most manufacturers do not disclose costs of EV models, nor do they share the expenses incurred for batteries. Automakers tend to

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What are they?	Why Do They Pose a Challenge?
declining costs of essential components like batteries. This inquiry is pivotal in determining the necessity of government subsidies going forward, especially considering automakers' recent EV price wars. Complementing this, another significant area of research is the effect of removing government financial incentives, such as exemptions for EVs from annual car taxes, on the used EV market.	keep such data confidential. Therefore, answering these questions necessitates the use of proxy datasets aggregated at a higher level. This type of data is seldom available in the public domain, and there are only a few companies that compile aggregated versions of these datasets, often charging substantial fees for access. To assess the impact of financial incentives on the used market, it will be necessary to obtain transaction data for the used car market or to conduct a survey on a representative sample of used car buyers. Conducting research on the strategic choices of EV startups, including managing supply chains for scaling up production, poses significant challenges due to the confidential nature of business strategies and procurement contracts. EV startups operate under intense competitive market pressures, which compel them to safeguard detailed strategic information. Additionally, these companies are often in their formative stages, during which their strategies and business models are constantly evolving. The absence of standardized data and limited collaboration between academia and industry further compound these challenges, as researchers may struggle to access detailed and reliable data consistently over time.
Theme 6: Challenges and Setbacks for EV Start-ups One emerging area of research interest is the examination of strategic choices made by these start-ups, which includes an analysis of their diverse business models, financing strategies, and approaches to collaboration. Such an investigation is crucial in identifying the factors that contribute to the success or failure of these ventures in a highly competitive market. Additionally, a critical challenge for EV start-ups lies in scaling-up production while efficiently managing global supply chains. This aspect requires a detailed inquiry into how these companies navigate the intricacies of large-scale manufacturing and the logistical demands of a global market.	Exploring the extent to which automakers can reduce vehicle prices is inherently challenging due to the opacity of cost structures and mark-ups. Regarding market entry strategies, forecasting sales in new markets is particularly complex. The success of a model in a new market depends heavily on consumer preference for that model, which can be unpredictable and vary significantly across different cultural and economic contexts. As for potential trade wars, the likelihood of such events and the strategies that countries might adopt depend on ongoing political negotiations, which are often confidential and dynamic. Researching these aspects requires expert elicitation studies, which themselves are limited by the availability and reliability of expert insights, as well as their inherent subjectivity.
Theme 7: Price War and Competition in the EV Market A key emerging area of research is the intensification of automakers' price wars aimed at increasing their EV market share in existing regions, coupled with strategic decisions to enter new markets. This aggressive competition raises questions about the long-term sustainability of such pricing strategies and the implications for global market dynamics. Equally important is the consideration of import tariffs designed to protect local automotive industries from the influx of foreign brands, particularly Chinese. The likelihood of such protective measures being adopted, their potential impact on the international trade and market equilibrium, and the possible backlash from China constitute critical areas for investigation.	The advanced battery technologies and products being developed by different companies often cannot be independently evaluated because these companies are highly concerned about sharing their technologies and products for such evaluations due to proprietary concerns, fearing the theft of their intellectual property. Without independent evaluation, understanding the limitations of these technologies and products is not straightforward. The development of software solutions that can facilitate communication between vehicles, chargers, and the energy grid, is challenged by the nascent stage of these technologies and the absence of
Theme 8: Emerging Technologies in EV Battery and Charging Solutions The development and scaling of advanced battery technologies for use in EVs represents a crucial emerging frontier, including the technological aspects of battery innovation. The second pivotal area lies in the potential for software and other technological solutions to optimize the use of EV charging infrastructure. This involves a multifaceted approach, where the focus is not just on enhancing the efficiency of charging systems but also on their seamless integration	

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What are they?	Why Do They Pose a Challenge?
with renewable energy sources and the electrical grid.	standardized protocols across different regions and manufacturers. Additionally, such research is hindered by insufficient collaboration among stakeholders—ranging from technology developers to utility companies and regulators—which is crucial for ensuring interoperability and effective integration of these systems.

Although the research priorities identified in this manuscript are derived from a global perspective, regional variations may influence their applicability due to differences in policy, economic conditions, technological adoption rates, and market maturity. For instance, the impact of U.S. and European policies on the domestic EV battery industry might differ significantly from those in Asian markets like China and India, where government incentives and supply chain dynamics vary. It is crucial for future studies to consider these regional differences to tailor strategies that address localized needs effectively.

4.2. Expert survey-based prioritization

Among the 19 participants,⁵ 16 represented non-profit organizations while the remaining respondents were affiliated with for-profit entities. In terms of education qualifications, 13 individuals possessed a doctoral degree, 5 individuals held a master's degree, and 1 individual possessed a bachelor's degree or lower. Regarding the duration of expertise possessed by professionals in the EV sector, a majority, over 50 % of the participants reported having accumulated 10 years of experience. On a scale ranging from 1 (very low familiarity) to 5 (very high familiarity), experts rated their familiarity with the research challenges examined in this survey as 3.21 on average. When asked to select all EV-related areas from their current and past work, 15 experts chose policy, 11 chose technology, 6 chose fuels, and 5 chose infrastructure.

We first determine the relative significance of three distinct criteria: importance in enabling the EV sector, novelty, and feasibility. Leveraging the AHP approach, we gathered insights from experts who performed pairwise comparisons among the mentioned criteria. The results from the AHP analysis, as shown in Fig. 2, reveal the following weights for each criterion :

Importance was deemed the most critical factor, capturing a weight of 58.8 %. The high weightage suggests that, among the three criteria, the perceived importance of a research challenge in enabling the EV sector played a paramount role in its prioritization by the respondents.

With a weight of 21.0 %, novelty emerges as the second most valued criterion. This indicates that while respondents value innovative research topics, the novelty of a topic is secondary to its overall importance in the context of the EV sector.

At 20.2 %, feasibility received the lowest weightage among the three criteria. This suggests that while the practicality of implementing or realizing a research topic is considered, it is not the primary driving factor for the experts.

Overall, the derived weights offer insights into the prioritization mindset of the experts. Importance to enabling the EV sector took precedence, underscoring its foundational role in decision-making. Meanwhile, while novelty and feasibility are considered, they play a supplementary role in the overall evaluation.

Furthermore, the consistency of the pairwise comparison matrix was assessed to ensure the reliability and validity of the results. The

⁵ The sample size of 19 experts, while relatively small, is not uncommon in the literature on expert surveys (James et al., 2022; Hulme et al., 2023; Gleser and Elbert, 2024; Robison et al., 2023; Moro et al., 2018).

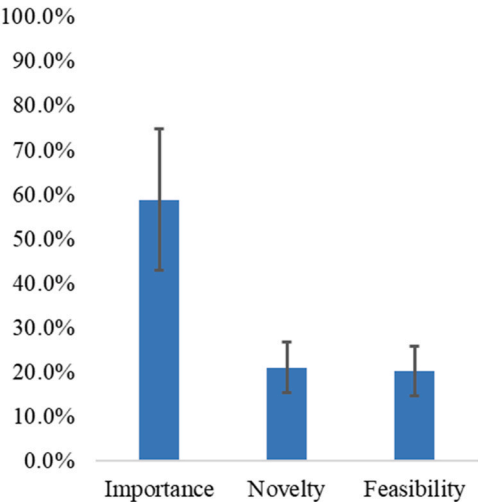


Fig. 2. AHP derived weights for the three criteria – importance in enabling the EV sector, novelty, and feasibility - determined based on the pairwise comparisons provided by the experts.

consistency ratio (CR), based on Eq. (9), was computed to be 8.2 %. As a rule of thumb, a CR value below 10 % is considered acceptable and indicates a reasonable level of consistency in the respondents’ evaluations.

In an endeavor to prioritize the identified emerging research challenges within the realm of the EV sector, we employed the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) approach on the survey data. This multi-criteria decision-making method facilitated the ranking of research alternatives by assessing their closeness to an ideal solution. We begin by normalizing the respondents’ ratings for the different alternatives on the three criteria, as shown in Table 1, followed by calculating the positive ideal solution, negative ideal solution and relative closeness to the ideal solution C_i^* , further reported in Table 1.

4.2.1. Sensitivity analysis

To account for sensitivity, we considered three unique respondent rating aggregation schemes for the TOPSIS analysis: (i) equal weighting - all respondents were given equal weight; (ii) experience-based weighting - weighting was determined based on the respondents’ experience; (iii) familiarity-based weighting - weighting was attributed according to the respondents’ familiarity with the topics. The rationale for selecting these three distinct weighting schemes lies in their potential to reveal

the nuances in prioritization that different respondent perspectives bring to our analysis. The ‘equal weighting’ scheme treats every expert opinion as equally valuable, providing a baseline that assumes all experts, regardless of background or exposure, contribute equally important insights into the prioritization process. The ‘experience-based weighting’ scheme, on the other hand, assigns greater influence to those respondents possessing more years of involvement or a higher degree of professional engagement within the EV sector. This approach acknowledges that depth of experience might affect the perception of what constitutes an important, novel, or feasible research challenge. Lastly, the ‘familiarity-based weighting’ scheme elevates the preferences of respondents who report a higher level of familiarity with the specific research challenges identified in our survey. This method posits that familiarity—potentially encapsulating knowledge of current trends, recent research, or ongoing industry developments—could shape one’s views on which research priorities are most critical. Together, these three weighting schemes were deployed to conduct a sensitivity analysis with the aim of examining how different assumptions about the weight of expert opinions could affect the prioritization of research challenges. This sensitivity analysis provides a means to test the robustness of our findings and to explore how alternative perspectives within our expert panel might shift the perceived importance of the research topics under consideration.

The resultant normalized TOPSIS scores ($norm_C_i^*$ in Eq. (8)), representing the closeness of each alternative to the ideal solution under the three aggregation schemes, are shown in Fig. 3(a)–(b). Furthermore, based on the normalized TOPSIS scores, we ranked the research alternatives, with the alternative closest to the ideal solution ranked first. Table 2 displays the corresponding rankings for the different alternatives based on the normalized TOPSIS scores, along with the sensitivities of these rankings to the chosen weighing scheme.

These rankings present a hierarchical understanding of the alternatives’ significance as perceived by the respondents. Interestingly, while certain alternatives consistently ranked high or low across the three aggregation schemes, some displayed variability, underscoring the influence of respondent weighting on the outcome. Overall, the results are fairly robust as 3 out of 8 alternatives encounter a maximum change of two ranks across three weighting schemes.

Across the three distinct respondent weighting schemes within the AHP-TOPSIS framework, the following emerging issues ranked among the highest considering importance in enabling the EV sector, novelty and feasibility aspects:

- Analyzing the effectiveness of U.S. and European policies in decreasing dependence on China for EV batteries and their effects on

Table 1
The decision matrix-TOPSIS, and relative closeness to the ideal solution scores for the case of equal weighting for all respondents.

Normalized decision matrix		Importance		Novelty		Feasibility	
Alt-1		0.38		0.38		0.33	
Alt-2		0.37		0.34		0.36	
Alt-3		0.36		0.35		0.31	
Alt-4		0.34		0.33		0.36	
Alt-5		0.35		0.35		0.39	
Alt-6		0.33		0.37		0.33	
Alt-7		0.34		0.33		0.39	
Alt-8		0.35		0.36		0.33	
Weighted normalized decision matrix		Importance	Novelty	Feasibility	S_i^+	S_i^-	C_i^*
Alt-1		0.22	0.08	0.07	0.012	0.028	0.71
Alt-2		0.22	0.07	0.07	0.011	0.024	0.67
Alt-3		0.21	0.07	0.06	0.021	0.016	0.43
Alt-4		0.20	0.07	0.07	0.024	0.013	0.35
Alt-5		0.21	0.07	0.08	0.017	0.021	0.55
Alt-6		0.20	0.08	0.07	0.028	0.010	0.27
Alt-7		0.20	0.07	0.08	0.023	0.018	0.44
Alt-8		0.21	0.08	0.07	0.020	0.013	0.40
A^+		0.22	0.08	0.08			
A^-		0.20	0.07	0.06			

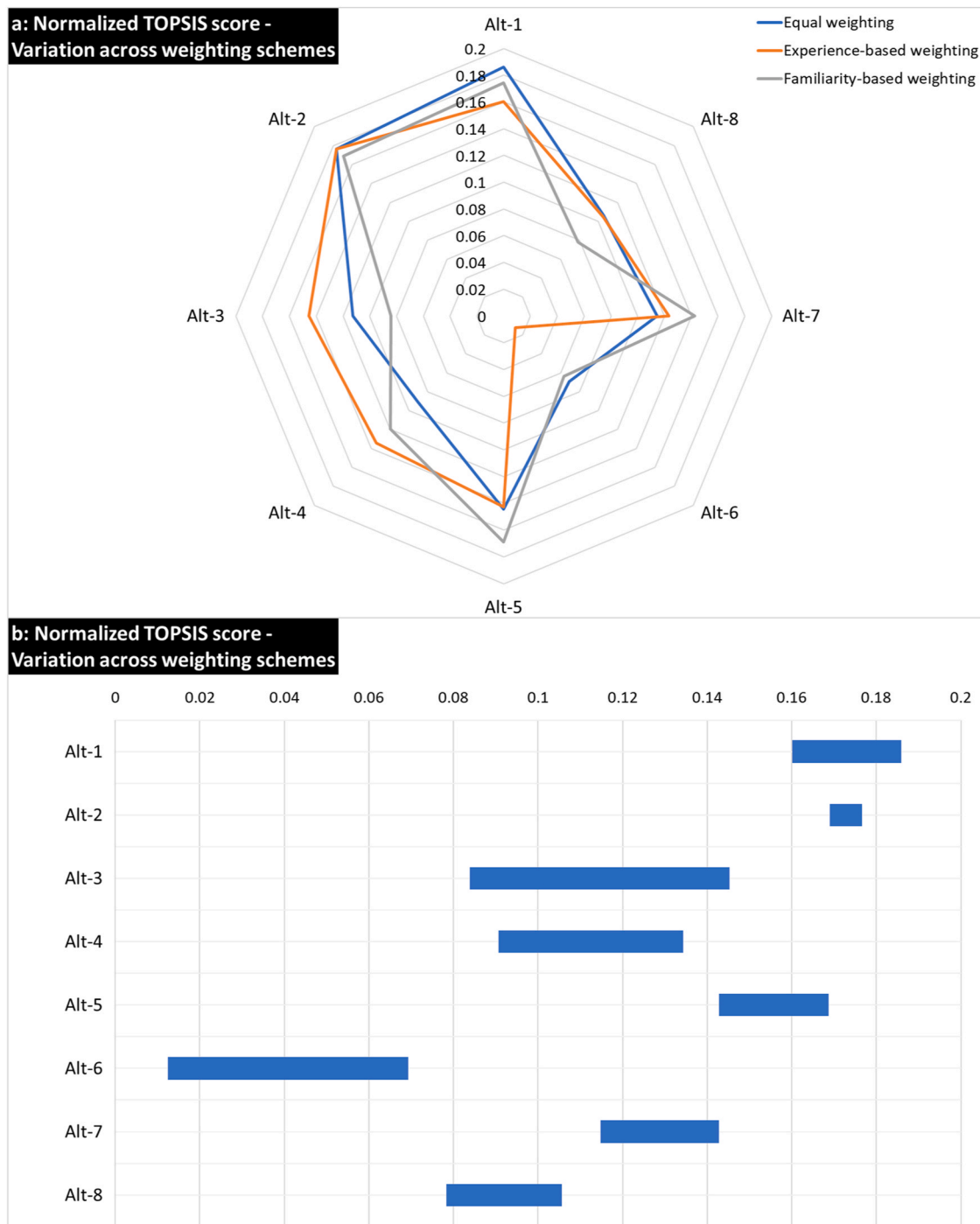


Fig. 3. (a) Normalized TOPSIS scores for potential research alternatives considering 3 cases of: (1) equal weights for all respondents, (2) respondent experience-based weights, (3) respondent familiarity-based weights, and (b) bars representing variation across the three weighting schemes.

the growth and competitiveness of the domestic battery industry, as well as international trade relations.

- Analyzing the influence of raw material prices on the cost of battery and EV production, and examining how this impacts EV sales, while also assessing the effect of car and battery makers' investment strategies and partnerships on the costs of securing these raw materials.
- Evaluate whether the reduction in electric car prices over time, influenced by falling battery costs and competitive pricing among automakers, still necessitates government subsidies, and examine the implications of removing these subsidies on the used electric vehicle market.

Among the various respondent weighting schemes employed within the AHP-TOPSIS framework, the following issues were found to have relatively low rankings:

- Examining the strategic choices and challenges faced by EV startups, encompassing their business models, financing strategies, collaboration tactics, as well as the issues related to scaling-up production and managing global supply chains.

Across the three distinct respondent weighting schemes within the AHP-TOPSIS framework, the following issues exhibited significantly divergent rankings:

Table 2

Expert ratings-based ranking of potential research challenges, considering 3 cases: (a) equal weights, (b) respondent experience-based weights, and (c) respondent familiarity-based weights. The rankings were obtained using the combined AHP-TOPSIS approach.

Alternatives	Equal weights	Experience based weights	Familiarity based weights
Alt-1: Governments' Strategies to Localize the EV Battery Supply Chain	1	2	1
- Assessing the effectiveness of U.S. and European policies in reducing reliance on China for EV batteries - Impact on domestic industry growth, competitiveness, and international trade relations			
Alt-2: EV Rollout Challenges: Rising Raw Material Prices and Automakers' Responses	2	1	2
- How raw material prices could affect the cost of battery and EV production and correspondingly EV sales - Impact of car and battery makers' investment plans and partnerships across the entire supply chain on costs of secured raw materials			
Alt-3: The Geopolitical Battle for Control of Critical Metals for EVs	5	3	6
- Cost-benefit analysis of OPEC-like group formation for battery metals, including impact on prices, supply chain stability, & market power - Impact of critical metal demand on supplier countries' economic growth & social welfare			
Alt-4: Battery Start-ups and Government Support	7	5	5
- Impact of governments' support in attracting investment for developing domestic battery industry & its competitiveness - Cost-benefit analysis of government funding for domestic battery industry including economic growth potential			
Alt-5: EV Pricing and Government Incentives: New and Used Vehicle Markets	3	4	3
- Have electric car prices fallen over time and do they still need government subsidies, especially with falling battery prices and EV price war among automakers? - Impact on used car EV market of removing government financial incentives, such as exemption for EVs from annual car tax			
Alt-6: Challenges and Setbacks for EV Start-ups	8	8	8
- Studying the strategic choices by EV startups, including their business models, financing strategies, collaboration strategies - Investigating challenges for start-ups related to scaling-up production and managing global supply chains			
Alt-7: Price War and Competition in the EV Market	4	6	4
- Automakers' price wars to increase EV market share in existing regions & decisions to enter new ones - Import tariffs to protect local automotive industry from foreign brands, particularly Chinese: Likelihood, impact & potential backlash			
Alt-8: Emerging Technologies in EV Battery and Charging Solutions	6	7	7
- Challenges & opportunities of developing & scaling advanced battery technologies for use in EVs - Potential for software & other technologies to optimize EV charging infrastructure use & integration with renewable energy sources & the grid.			

- Analysis of the impact of critical metal demand on supplier countries' economic growth and social welfare, along with a cost-benefit analysis of these countries forming an OPEC-like group for battery metals, examining the implications for prices, supply chain stability, and market power.
- The dynamics and consequences of automakers engaging in price wars to expand their electric vehicle (EV) market share in existing regions and their strategies to enter new markets, in conjunction with the likelihood, impact, and potential backlash of countries imposing import tariffs to protect their local automotive industries from foreign competition, especially from Chinese brands.

5. Conclusions

This work systematically explores the multifaceted challenges and emerging research priorities pivotal for enabling the EV sector. By delving into diverse themes, it provides a comprehensive analysis of the current state and future directions of research.

By employing three distinct respondent weighting schemes within the AHP-TOPSIS framework, we identify key emerging issues in enabling the EV sector that are critical in terms of importance, novelty, and feasibility. Among the top three ranked research challenges that merit in-depth exploration include – (i) U.S. and European policies impact on EV battery industry and trade relations, (ii) rising metal prices' impact on battery and EV markets; (iii) EV price trends and whether government subsidies are still needed given falling battery costs and EV

price war among automakers. Within the AHP-TOPSIS framework, our sensitivity analysis employed three different response weighting schemes, revealing a significant variability in expert opinions-based ranking for two issues: the geopolitical struggle for dominance over critical metals essential for EV batteries, and the intense price competition within the EV market.

While this study offers a broad overview of the issues at hand, there are certain limitations. The rapidly evolving nature of emerging issues in the EV sector implies that findings might require continual updates. In addition, it is possible that there may be blind spots or specific issues that have not yet been reported in the media, and thus remain unaddressed in this work. Furthermore, the process of transforming contemporary issues into emerging research prospects is inherently subjective, which suggests that alternative perspectives could lead to the identification of distinct emerging research opportunities related to these issues. Moreover, the reliance on expert inputs, although valuable, may not fully capture the diversity of perspectives in this global industry.

This work contributes to the field by offering a structured framework to analyze and prioritize research areas in the EV sector. It bridges the gap between academic discourse and practical application. In particular, the delineated priorities could serve as a potential roadmap for guiding forthcoming research endeavors within both the industrial and academic domains. Research findings from such initiatives may yield practical recommendations for policymakers and industry stakeholders, consequently fostering the formulation of strategies aimed at expediting

the emergence of a resilient EV sector. In conclusion, the emerging issues identified through journalistic reports, a review of published literature, and expert consultations provide a solid foundation for identifying the imperative research priorities required to facilitate a robust EV sector.

CRedit authorship contribution statement

Saif Almutairi: Writing – review & editing, Methodology, Investigation, Conceptualization. **Prateek Bansal:** Writing – review & editing, Methodology, Conceptualization. **Rubal Dua:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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